

# Skills Summary

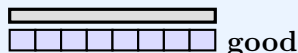
## Gaps and Overlaps When We Measure

*Use this reference while completing your worksheet or when studying.*

### Skill 1 — Lay the cubes with no gaps and no overlaps

#### Rule:

Start at one end. Put the cubes **side by side**: touching, but never on top of each other, and never with a space between. Then the **number of cubes is the length**.



**Example:** A pencil is covered by cubes laid end to end, with no gaps and no overlaps. How long is the pencil?

**Step 1.** The cubes touch and none sit on top of another, so every cube covers new pencil — that means I may simply count the cubes.

**Step 2.** Touch each cube once and count on without skipping or repeating; the last cube counted is 6.  $\Rightarrow$  the pencil is **6** cubes long

**Try it:** A spoon is covered by cubes laid end to end, no gaps and no overlaps, and the last cube counted is 4. How long is the spoon?

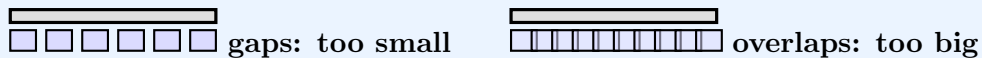
check: 4

### Skill 2 — Spot a gap or an overlap mistake

#### Rule:

**Gaps** leave bare spaces, so too few cubes reach the end: the count is **too small**.

**Overlaps** let cubes pile onto each other, so extra cubes squeeze on: the count is **too big**.



**Example:** Ben left gaps and counted 5. A careful measure gives 7. Write  $<$  or  $>$ : Ben's count \_\_\_\_\_ real length.

**Step 1.** Ben left gaps, so I expect his number to be too small — but I still check by comparing the two counts.

**Step 2.** Ben's count is 5 and the real length is 7; 5 comes first when we count, so it is the smaller number, and the symbol opens toward the bigger one.  $\Rightarrow$  Ben's count  $<$  real length

**Try it:** Lia let her cubes overlap and counted 9. A careful measure gives 6. Write  $<$  or  $>$ : Lia's count \_\_\_\_\_ real length.

check:  $<$

### Skill 3 — Fix the count after a gap or overlap

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**Rule:**

The object never changes. Only the lay-down does. To fix a count, take the **careful** length and the **sloppy** count and find **how far apart** they are — count on from the smaller number to the bigger one, and that is how many cubes the gaps hid or the overlaps added.

**Example:** Ben's gaps: he counted 5, but the careful length is 7. How many cubes did his gaps hide?

**Step 1.** "How many hidden" asks how far apart the two numbers are, so this is a difference, not a total.

**Step 2.** Count on from 5 up to 7 — that is two counts, so the difference is 2.  $\Rightarrow$  the gaps hid **2** cubes

**Try it:** Lia's overlaps: she counted 9, but the careful length is 6. How many extra cubes did the overlapping add?

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(!) **Watch out:** A *bigger number* is not a *better measurement*. Always look at how the cubes were laid before you trust a count — and never add the two counts when a question asks "how many more" or "how many hidden".